

Q&A Prep — Group 13, AI-623 DVLM Final Presentation

Last updated: 2026-05-30 (day before submission). 30-mark Group Q&A plus 30-mark Individual Q&A = **60 marks ride on the answers below.**

Section ownership (suggested split for Individual Q&A)

Member	Owns
Eeman Adnan	§1 Introduction, §2 Related Work, §6 Discussion, broader-claims and motivation questions
Muhammad Aslam Adnan	§3 Methodology, §4 Experimental Design, §5.1 main grid, §5.2 motion, §5.3 baselines, implementation details
Mufaddal Hatim	§5.4 posterior diversity, §5.5 significance, §5.6 ablations, paradoxical reversal, 8 failed ablations, statistical methodology

For Group Q&A, the member who owns the section takes the lead; others add supporting points.

A. Problem and motivation (Eeman lead)

A1. Why image deblurring, and why CelebA-HQ-256 faces? - Deblurring is the canonical Bucket-3 inverse problem ($y = Ax + \epsilon$) with a known operator and well-understood baselines - Faces are a controlled domain where both DPS and FlowDPS priors have pretrained checkpoints at matched 256×256 resolution - Allows clean comparison without confounders from dataset distribution shift; the limitation is documented in §6 and is the natural future-work direction (LSUN-Church, DIV2K)

A2. What was the original proposal, and did you stick to it? - Proposed in March: compare DPS vs FlowDPS quality, efficiency, robustness, and against classical baselines - Stuck to all four mandatory comparisons; **added three of the four Bucket-3 stress-test axes** (operator mismatch, posterior diversity, gradient-free vs gradient-based) when only one was required - The headline finding contradicted the proposal’s hypothesis (“FlowDPS within 0.5–1.1 dB of DPS at 2–5× speedup”). We report the contradiction honestly rather than reshape goals

A3. Why is this inverse-problem setup important? - Real applications: medical imaging (MRI reconstruction), astronomy (telescope deconvolution), computational photography - Each cares about a slightly different point in the PSNR-vs-uncertainty trade-off our work characterizes

A4. What’s the recent shift the field is making? - Toward flow-based posterior samplers, blind/semi-blind operators, and gradient-free formulations - Our work directly tests the first claim (FlowDPS quality) and the third (gradient-free with particle_dps + tempered SMC)

B. Methodology (Aslam lead)

B1. What’s the core difference between DPS and FlowDPS-style guidance? - DPS: stochastic DDIM reverse process + likelihood-gradient correction. Uses Tweedie’s identity for $\hat{x}_0$ - FlowDPS: deterministic Euler

ODE integration of a velocity field + the same form of likelihood-gradient correction on the Tweedie-clean-image identity for rectified flow ($\hat{z}_1 = x_t + (1-t) v_{\theta}$) - Mathematically identical update structure; differs in (a) the prior's generative process and (b) the temporal-noise schedule

B2. Why did you reimplement FlowDPS? Was that risky? - The official Kim 2025 code (github.com/jeongsol-kim/flowdps) is locked to Stable Diffusion 3 latent space at 768×768 - Without reimplementation, a head-to-head DPS-vs-FlowDPS comparison would span different resolutions, datasets, and prior architectures — the gap would be ambiguous - Risk: implementation bugs could produce false underperformance. We mitigated by (i) sanity-checking unconditional generation (matches Liu 2023 reference samples), (ii) verifying L2-norm vs L2-squared convention against published code, (iii) diagnosing the EMA bug

B3. What's the EMA bug, and how does it affect results? - The Liu 2023 RF checkpoint has 644 EMA shadow params for a 645-parameter network - The missing one is `all_modules.0.W` (Gaussian Fourier projection), registered with `requires_grad=False`, silently skipped by the gno-bitab EMA tracker - Naive EMA loading randomly initializes the Fourier projection → 7.76 dB PSNR collapse - Correct procedure aligns the 644 shadow params to trainable params AND copies `all_modules.0.W` from the raw `model` state-dict - This fix is the 1st confirmed ablation win (`flowdps_rf_v2`, +0.77 dB) - Even our v1 baseline (used throughout the headline numbers) does NOT apply this fix — it uses the raw `model` state-dict directly, matching what a naive reimplementer would write

B4. Why DDIM and not DDPM sampling for Pixel-DPS? - DDIM has fewer NFE for the same image quality; it's the standard for guided sampling in the literature - The diffusers DDIMScheduler is well-tested and pinned at 0.30.1

C. Experimental design (Aslam lead)

C1. Why 50 images per cell? Isn't that small? - 50 is large enough for paired t-tests to detect Cohen's $d > 0.5$ at $\alpha=0.05$ with >80% power — the effects we care about have $d > 1$ - For the borderline `flowdps_rf_spectral` matched grid cells, we explicitly increased to $n=100$ because t-stats were near the Bonferroni threshold — this was a sample-size augmentation, NOT hyperparameter tuning (no spectral parameters changed) - We did not retune to chase significance — that would be p-hacking

C2. Why Bonferroni and not a softer correction (e.g., FDR)? - Bonferroni is the conservative choice; passing it means we're confident even by the strictest standard - A softer correction would have promoted more ablations, but we'd rather underclaim than overclaim - Two near-misses we report (`pixel_dps_sched_v2`, `flowdps_rf_heun`) would pass a permissive criterion; we explicitly call them near-misses

C3. What's the seed protocol? - Per-image seeds drawn from a fixed schedule make per-cell paired comparisons deterministic - The same seeds are used across methods for the paired t-test to be valid (matched samples differ only by method, not by image or noise realization)

D. Main results (Aslam lead, supported by Mufaddal)

D1. What's the headline? - Pixel-DPS dominates FlowDPS-on-RF on matched Gaussian (+1.4 to +4.2 dB across 6 cells, all Bonferroni-significant) - Same ordering on matched motion (+3.95 dB mean across 12 cells, all significant, slightly wider gap) - Reverses under operator mismatch (FlowDPS-on-RF degrades less) - Pixel-DPS produces a sharp point estimate; FlowDPS-on-RF produces a 3.5× broader posterior

D2. Why does FlowDPS-on-RF saturate by NFE=25? - Working hypothesis: the rectified-flow velocity field has more “stiff” dynamics late in the trajectory than DDPM's noise-prediction. Once the trajectory has flowed toward

the data manifold, additional Euler steps move very little - Empirically confirmed: from NFE=25 to NFE=200, FlowDPS-on-RF gains <0.5 dB; Pixel-DPS gains 1–2 dB - This diagnosis motivated Tier-A’s ramp- ζ schedule (more guidance at late steps), which produced the 1st confirmed win

D3. DnCNN+PnP beats your FlowDPS on 5/6 noisy cells. Does that mean your FlowDPS is broken? - No — it means DnCNN+PnP is a strong baseline that we should respect (the §5.3 “Why DnCNN+PnP beats FlowDPS-on-RF” paragraph in the detailed report covers this) - DnCNN is a 668k-parameter bias-free denoiser specifically trained on Gaussian noise residuals (the prior task) at varied σ ; inside PnP-ADMM it contributes the right prior at near-zero per-iteration cost - FlowDPS-on-RF uses a 99M-parameter velocity field for 100 NFE plus backprop at every step — but the Liu 2023 RF prior is undertrained relative to modern large diffusion priors. The gap is a model-capacity issue, not a sampler-algorithm issue - Our Tier-A win (`flowdps_rf_v2`) recovers +0.77 dB without changing the model — confirming the sampler is correct and the gap is mostly about the v1 zeta+EMA tuning

E. Operator-mismatch reversal (Mufaddal lead with Aslam)

E1. Why does the comparison reverse under mismatch? - Hypothesis 1: the deterministic ODE acts as implicit regularization that’s robust to wrong gradient direction - Hypothesis 2: FlowDPS’s broader posterior (3.5 $\times$) already encodes more uncertainty, so a wrong-operator gradient gets “averaged out” more - Hypothesis 3: Pixel-DPS’s L2 gradient is more aggressive per-step (larger trust-region) and thus more brittle to misspecification - We don’t have a definitive answer; we report the empirical reversal honestly. The §6 Discussion paragraph “Gradient-based vs gradient-free guidance” speculates further

E2. Why is this finding important? - Implies the choice between DPS and FlowDPS is application-dependent, not universal - Calibrated operator $\rightarrow$ DPS. Uncertain operator (medical imaging with patient-varying PSF, telescope deconvolution with atmospheric blur) $\rightarrow$ consider flow-based posterior sampling

F. Posterior diversity (Mufaddal lead)

F1. What does “PSNR $\neq$ posterior” mean concretely? - A single sample’s PSNR tells you how close one reconstruction is to ground truth — it’s a point estimate - A well-calibrated posterior should have non-zero variance reflecting the true ambiguity in the measurement - Pixel-DPS samples differ by LPIPS 0.05 across seeds (essentially identical) — it’s collapsed to a single modal estimate - FlowDPS-on-RF samples differ by LPIPS 0.175 — visibly different hair, skin shading, micro-features (Figure 2)

F2. Is this Bayesian-vs-frequentist? - More like point-estimate vs posterior. Both methods are sampling from approximations to the same posterior; the difference is the variance of the approximation - Pixel-DPS’s L2 gradient is sharper, pushing samples toward a single mode. FlowDPS-on-RF’s deterministic trajectory + flow-based prior preserves more sample-to-sample variation

F3. Why does FlowDPS-RF benefit more from posterior averaging? - Mean-of-N PSNR gain: Pixel-DPS +0.89 dB, FlowDPS-RF +2.15 dB - Bayesian gain from averaging independent posterior samples is large when individual samples are noisy and approximately unbiased - FlowDPS-RF samples are noisier (more diverse) but unbiased; Pixel-DPS samples are deterministic (no Bayesian gain available)

G. Ablation methodology (Mufaddal lead)

G1. Why 13 ablations? Isn't that fishing? - All 13 were planned in 4 tiers (A/B/C plus rounds 2/3/4/5) before measurement, motivated by specific diagnoses from the v1 baselines (saturation curve, robustness failure, etc.) - We used a strict Bonferroni-corrected promotion criterion to control multiple-comparison inflation - 8 of 13 failed the strict gate; we report them all honestly with mechanistic diagnoses, not just the 3 wins - A “fishing” approach would have reported only the wins and not documented the failures

G2. Why a strict per-cell gate, not aggregate Δ ? - Aggregate Δ can be pushed positive by 1–2 strongly-positive cells while 4+ cells are flat or negative - Per-cell + Bonferroni demands the effect to be broadly present - 5 of our 13 candidates would have passed an aggregate- Δ criterion but failed the per-cell gate — they're documented as near-misses

G3. The 3rd win required $n=100$. Wasn't that p-hacking? - The hyperparameters were fixed before the $n=50$ measurement; $n=50$ produced 3/6 cells significant with the $\sigma_b=5$ cells at $t \approx 2.5$ — above the single-test critical value but below the Bonferroni-corrected one - We doubled n to 100 to reduce standard error (statistical power augmentation, the textbook correct response to a borderline comparison) - **No spectral hyperparameters changed** between $n=50$ and $n=100$. The ϵ , α , and kernel weighting are identical - If we had tuned the spectral parameters to push past $p < 0.05$, that would be p-hacking. We did not

H. The paradoxical reversal (Mufaddal lead — most likely Q&A target)

H1. What's the most interesting finding of the whole project? - The Tier-B spectral-weight mechanism has *opposite* verdicts in adjacent regimes: - Operator mismatch $\rightarrow$ fails (-0.18 dB Pixel-DPS, -0.20 dB FlowDPS-RF) - Matched conditions $\rightarrow$ wins ($+0.46$ dB Pixel-DPS, $+0.54$ dB FlowDPS-RF) - Confirmed on **both** posterior samplers (algorithm-agnostic) - Mechanistically interpretable: downweight residuals where the *assumed* operator's spectrum is noise-dominated. Under matched conditions, those are the right bands. Under mismatch, they are the wrong bands

H2. Why is this publishable? - A brittle ablation would not produce a clean reversal — it would fail in both regimes or succeed in both - The reversal is evidence that the mechanism is correctly diagnosed - Algorithm-agnostic (works on both DPS and FlowDPS-style guidance) - Workshop-pitchable as a stand-alone finding

H3. Can you explain the mechanism more deeply? - Spectral weight: $W(f) = 1 / (1 + \alpha \cdot \text{ReLU}(\epsilon - |H(f)|))$ - $|H(f)|$ is the assumed forward operator's spectral magnitude - Below noise threshold ϵ , the weight increases; that frequency's residual is *downweighted* in the DPS gradient - Intuition: at those frequencies, the assumed operator can't see signal anyway, so residuals there are mostly noise. Downweighting prevents the sampler from being misled by them - Under matched conditions, the assumed operator's noise-dominated bands are the *true* operator's too. Win. - Under mismatch (motion measurement, Gaussian assumed): the assumed operator says “frequencies above some cutoff are dead.” But the true motion-blur operator preserves directional content that the assumed Gaussian cannot see. Downweighting those bands throws away the actual signal. Loss.

I. Failed ablations (Mufaddal lead)

I1. Why did the Π -GDM tier fail so badly (-13.6 dB)? - L2-norm vs L2-squared convention mismatch - DPS uses $\|y - A(\hat{x}_0)\|$ (L2 norm); Π -GDM uses $\|y - A(\hat{x}_0)\|^2$ (L2 squared) with a different leading factor ($\frac{1}{2}\sigma_y^2$) - Conflating them gives a wrong leading factor that depends on the σ_y scale, blowing up the gradient at noisy cells - The corrected analytical Π -GDM (`flowdps_rf_pigdm_pure`) reduces the margin from -9.8 to -3.4 dB, consistent with the literature finding that Π -GDM underperforms L2-DPS on Gaussian blur on this prior

I2. Why did particle DPS fail catastrophically? - Path degeneracy: at $P=8$ particles, the effective sample size $P_{\text{eff}} < 1.5$ within ~ 10 reverse steps of the diffusion trajectory - The unguided DDIM step separates particles by more than the measurement likelihood's effective σ_y ; importance weights become near-singular; one particle dominates; the rest of the trajectory is identical to unguided DDIM on that one survivor - Classical Doucet 2001 failure mode; we didn't invent it - Fix would require either larger P ($P=64+$) or a non-trivial MCMC rejuvenation kernel (e.g., Langevin with the score). Out of scope for this project

I3. Why did Heun at NFE=200 fail? - Heun is 2nd-order accurate vs Euler's 1st-order. Karras 2022 EDM argues higher-order integrators help unconditional sampling - Our finding: Heun at NFE=200 (its natural budget) is -0.17 dB vs Euler+v2@NFE=100 (matched budget): a regression - Diagnosis: ODE integration accuracy is not the FlowDPS-on-RF bottleneck; **guidance saturation is**. v2's EMA + ramp- ζ already extracts the available signal; a better integrator on top adds nothing - Useful caveat to the EDM analysis when extended to guided sampling

J. Implementation and infrastructure (Aslam lead)

J1. What software stack? - PyTorch 2.4.1 + diffusers 0.30.1 (pinned), torchvision, LPIPS, Wiener-filter via scipy.fft - All Bonferroni / paired-t / Wilcoxon via scipy.stats - RTX 3060 12 GB, 32 GB RAM, single GPU

J2. How long did the experiments take to run? - Main grid: ~ 6 GPU-hours per method - Robustness study: ~ 4 GPU-hours - Each ablation grid: ~ 3 – 6 GPU-hours - Matched motion (most recent): ~ 7 GPU-hours - Total: ~ 50 – 60 GPU-hours of measurement time. All resume-safe via per-row CSV append+fsync

J3. Where can graders verify the numbers? - Public repo: https://github.com/adnan821/dps_flowdps - Every PSNR in the report traces to a row in

`outputs/results/{main_grid,robustness,baselines,motion_matched}.csv` - Lab notebook EXP-001..EXP-026 documents every decision - ABLATION_SCORECARD.md is the canonical wins/fails reference

K. Limitations and future work (Eeman lead)

K1. Your most honest limitation? - Single dataset (CelebA-HQ-256 faces). Generalization to LSUN-Church or DIV2K untested. Pixel-DPS's dominance could be face-specific - Two operator families (Gaussian + motion) on the matched comparison; broader operator classes (super-resolution, inpainting) untested

K2. What would you do with more time? - Multi-dataset replication (top priority — closes the most consequential limitation) - Implement DDRM correctly so we have a third learned-prior baseline - Scale gradient-free SMC to $P=64$ with MCMC rejuvenation kernel - Test broader operator classes (SR, inpainting) on the matched comparison

K3. Is this work publishable? - Workshop: plausibly, the paradoxical-reversal finding is the strongest single contribution - Top venue: not as is — would need either multi-dataset, an algorithmic novelty that wins on absolute PSNR (we narrow the gap but don't beat Pixel-DPS), or a deeper theoretical contribution

L. Likely attack questions (be ready for any)

L1. The proposal hypothesis was wrong. Didn't you change your goals? - We stuck to the same four research questions. The proposal *claim* (FlowDPS within 0.5 – 1.1 dB at 2 – $5\times$ speedup) was tested and falsified with quantitative evidence. That's a valid scientific outcome - Our headline conclusion remains the same shape: we now have a clear guideline for *when* to use each method

L2. Cohen's $d > 3$ is huge. Is your experimental noise too low? - Both samplers are deterministic given seed (Pixel-DPS up to floating point); paired comparisons across the same images have very low within-pair variance - Cell-to-cell variance is dominated by *image* difficulty, not seed noise. Cohen's d on paired differences measures effect size relative to within-pair noise, which is small by design - The same protocol applied to comparable studies in the literature produces similarly large d for established differences

L3. You implemented DDRM and it didn't work. Doesn't that question your reimplementation skills generally? - The DDRM SVD computation requires an explicit SVD of the operator, which is non-trivial for the Gaussian-blur+downsampling regime we attempted. We ran out of time to debug - DnCNN+PnP-ADMM works correctly and is also a learned-prior baseline - The FlowDPS reimplementation has multiple sanity checks (matches Liu 2023 unconditional samples; v1 baseline matches the literature qualitative behavior of saturation)

L4. Why didn't you use SD3 as the prior since that's what Kim 2025 uses? - SD3 operates at 768×768 in latent space; the matched DDPM (Pixel-DPS's prior) is at 256×256 in pixel space. Comparing across these would conflate prior+representation+resolution differences - A pixel-space reimplementation isolates the sampler choice as the only difference

L5. Your ablation methodology rejects 5 of 13 candidates that would pass a softer criterion. Isn't that overly strict? - We pre-registered the strict gate before measurement; it is the documented promotion criterion for the entire study - Methods that fail the strict gate but pass softer criteria are *reported* as near-misses, with the per-cell breakdown — readers can apply their own threshold - Strict gate is a rigor improvement, not a methodological weakness

L6. Your DnCNN+PnP baseline beats your FlowDPS reimplementation on 5/6 noisy cells. Doesn't that invalidate your FlowDPS? - It doesn't — DnCNN is a strong learned-prior baseline; that's a separate fact about the strength of supervised denoising priors - Our FlowDPS-on-RF is the correct implementation of the Kim 2025 algorithm on a smaller, less-trained RF prior (Liu 2023) - The Tier-A win (`flowdps_rf_v2` , EMA + ramp- ζ) closes the gap to DnCNN+PnP partially without changing the model, confirming the sampler is correct and the gap is model-capacity-related

M. Closing questions and pivots

M1. What's the one-line takeaway? - "Pixel-DPS dominates FlowDPS-on-RF on matched conditions by 1.4–4.2 dB and the matched-motion confirms this isn't Gaussian-specific. The ordering reverses under operator mismatch. Pixel-DPS is a sharp point estimate; FlowDPS-on-RF is a $3.5\times$ broader posterior."

M2. What's the one finding you're most proud of? - "The paradoxical reversal: the same spectral-weight algorithm fails under operator mismatch and wins under matched conditions, confirmed on both samplers. Algorithm-agnostic, mechanistically interpretable, publishable on its own."

M3. If you redid the project, what would you do differently? - Two datasets from the start (CelebA-HQ + LSUN-Church) — would have let us claim the matched-Gaussian dominance is general, not face-specific - Implement DDRM as the priority-1 baseline before FlowDPS-on-RF, since DDRM is the closer analog to Pixel-DPS than DnCNN+PnP - Pre-register a learned ζ schedule for FlowDPS-RF rather than discovering the saturation diagnosis post-hoc

Practice tips

1. **Do at least one full dry-run** with all three members reading their owned sections — time it (target ≤ 15 minutes for the prepared presentation, leaving 15 minutes for Q&A)
2. **Each member should read this document end-to-end** so cross-section questions can be passed to the right person fluently
3. **For Individual Q&A:** graders will likely probe each member on their owned section. Be ready to explain the *why* behind each choice, not just the *what*
4. **For Group Q&A:** practice handoffs (“Mufaddal can speak to that”) — looks professional and gets you full marks for cohesion
5. **Don’t bluff:** if you don’t know the answer to an attack question, say “I don’t know — here’s what we’d need to test to find out.” That’s worth more marks than a wrong confident answer